MINJUN KIM

Selected Technical Portfolio | Document Intelligence, Trustworthy AI, and Systems

RESEARCH	ENGINEERING	HUMAN-CENTERED AI
Evidence extraction Calibration Evaluation	Python / PyTorch Swift / macOS Distributed systems	Review workflows Accessible explanations User studies

PORTFOLIO OVERVIEW

This portfolio presents three fictional sample projects aligned with Minjun Kim’s graduate interests. Metrics, organizations, and links are illustrative and internally consistent with the accompanying application package.

PROJECT	ROLE	CORE OUTCOME
ApplyWise	ML + Product Lead	34% faster checklist completion
EvidenceLens	Undergraduate Researcher	Extraction F1: 0.84 to 0.91
EdgeVision	Systems Engineer	2.4x faster on-device inference

01 - APPLYWISE

Evidence-grounded application preparation agent | 2026

PROBLEM

Applicants must interpret long eligibility notices, compare them with personal records, track missing evidence, and format files correctly. Existing tools often summarize requirements but do not connect decisions to exact source passages.

SYSTEM

STAGE	IMPLEMENTATION	OUTPUT
Ingest	OCR + layout parsing	Structured notice and evidence
Match	Requirement-evidence graph	Eligible / ineligible / verify
Explain	Span-level provenance	Highlighted supporting passages
Prepare	Local macOS workflow	Renamed, converted, compressed files

CONTRIBUTION

- Designed the requirement-evidence schema and confidence policy.
- Implemented layout-aware extraction and evidence retrieval in Python and PyTorch.
- Built an evaluation set of 480 requirement-document pairs and led usability testing.

RESULTS

METRIC	BASELINE	APPLYWISE
Checklist completion time	18.2 min	12.0 min
Requirement matching accuracy	82%	94%
Unsupported eligibility claims	11%	3%

Lesson: explanations were most useful when they exposed both supporting evidence and unresolved uncertainty.

02 - EVIDENCELENS

Calibrated extraction for long-form administrative documents | 2024-2026

RESEARCH QUESTION

Can a compact multimodal model extract fields from heterogeneous documents while providing calibrated, reviewable evidence?

METHOD

- Combined OCR tokens with normalized bounding-box coordinates and visual region embeddings.
- Used document-family-aware splits to prevent template leakage.
- Calibrated span confidence with temperature scaling and evaluated expected calibration error.
- Created a reviewer interface that grouped low-confidence spans by error type.

EVALUATION

MODEL	FIELD F1	ECE	LATENCY / PAGE
Text-only baseline	0.84	0.142	0.41 s
Layout-aware model	0.91	0.061	0.56 s
Layout-aware + calibration	0.91	0.028	0.58 s

OUTCOME

The work produced a fictional workshop paper, a reproducible evaluation harness, and a design principle: uncertainty should be attached to evidence that a reviewer can inspect, not hidden behind a single document-level score.

03 - EDGEVISION

Efficient on-device visual inspection | 2025

GOAL

Deploy a defect-detection model on Apple Silicon for privacy-preserving offline inspection without materially reducing accuracy.

OPTIMIZATION PIPELINE

STEP	CHANGE	EFFECT
1	Profile preprocessing and memory copies	Removed redundant image conversions
2	Structured channel pruning	Reduced model size by 41%
3	Knowledge distillation	Recovered 2.7 points of accuracy
4	Core ML conversion and quantization	Enabled fast local inference

RESULT

METRIC	ORIGINAL	OPTIMIZED
Accuracy	94.8%	94.1%
Median latency	184 ms	77 ms
Model size	86 MB	34 MB

REFLECTION

This project taught me to treat efficiency as a full-system property. The largest gains came from profiling the data path and deployment runtime, not simply shrinking the neural network.

CONTACT

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